UNIVERSITY OF GRAZ
Department of Mathematics
and Scientific Computing

MAT.B0204UB MATHEMATICAL MODELLING
SUMMER SEMESTER 2025/26

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EXERCISE SHEET 2

Due: 15 April 2026, 10:00

EXERCISE 2.1 (From ex. sheet 1)
Perform the heuristic derivation to prove that the chemical reaction

$$A \quad + \quad B \quad \longrightarrow \quad C,$$

with rate constant $\lambda > 0$, is described by the following system of ODEs

$$\begin{cases} \dot{a} = -\lambda ab, \\ \dot{b} = -\lambda ab, \\ \dot{c} = \lambda ab. \end{cases}$$

Hint: bring inspiration from the heuristic derivation performed during the class for the linear case. More specifically, assume that particles of type A and particles of type B are (uniformly) distributed among N cells. Then introduce appropriate assumptions to determine the probability that the two different types of particles interact with each other within a single cell.

EXERCISE 2.2 *(A logistic equation with harvesting)*
Consider the following equation,

$$\dot{x} = x(1-x) - g_{\mu,a}(x), \qquad \text{with} \quad g_{\mu,a}(x) := \frac{\mu x}{1 + \frac{\mu x}{a}}, \tag{1}$$

and $\mu, a > 0$.

1. Perform the stability and bifurcation analysis of (1). Namely, determine the critical value μ_c and identify the type of bifurcation, depending on the parameters μ_c and a. Then, interpret the obtained result.
2. Consider a general function $g_{\mu,a}$. Determine sufficient assumptions for $g_{\mu,a}$ so that the same scenario studied in item 1, with $g_{\mu,a}$ as in (1), holds.

EXERCISE 2.3
Consider the following normal form for $\mu \in \mathbb{R}$ and $\varepsilon \in \mathbb{R}$, $|\varepsilon| \ll 1$,

$$\dot{x} = \mu x - x^3 + \varepsilon. \tag{2}$$

1. Perform the stability and bifurcation analysis of (2).
2. Interpret the results. Namely, interpret the persistence of the bifurcation phenomenon for $\varepsilon \neq 0$, if it occurs. Then, compare with the case $\varepsilon = 0$.